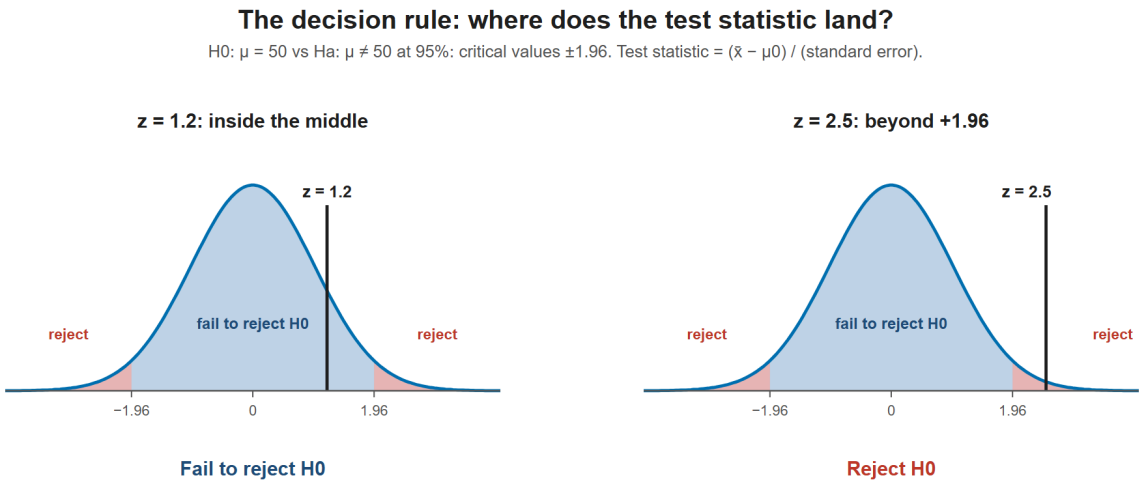
Lesson 10.4 - Test Statistic for Means and Population Proportions

> **Video:** Test Statistic For Means and Population Proportions by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

The decision rule

1. Find the **critical value(s)** from the significance level and the type of test. For a two-tailed test at 95%, ± 1.96.
2. Compute the **test statistic** z_c (or t) from the sample.
3. Statistic in the rejection region → **reject** H_0. Statistic in the fail-to-reject region → **fail to reject** H_0.

Example: $H_0 : \mu = 50$ vs $H_a : \mu \neq 50$ at 95% has critical values ± 1.96. A calculated z_c beyond 1.96 (or below −1.96) means reject.



Compute the test statistic, then see which region it lands in. Between −1.96 and 1.96, fail to reject H0. Beyond either critical value, reject H0.

Test statistic for a mean

Sample size	σ known?	Distribution	Test statistic
$n < 30$	**No** (only s)	**t**, df $= n - 1$	$t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
$n < 30$	Yes	z	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
$n \geq 30$	No (use s)	z (approximately normal)	$z = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
$n \geq 30$	Yes	z	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

Here μ is the value claimed in H_0.

> Many textbooks use the t-distribution **whenever σ is unknown**, whatever the sample size. For $n \geq 30$, t and z give nearly the same answer, so the video's rule is a common shortcut.

Test statistic for a proportion

$$z = \frac{\hat{p} - p}{\sqrt{\dfrac{pq}{n}}}, \qquad q = 1 - p$$

- $\hat{p}$ = sample proportion (x/n)
- p = the proportion claimed in H_0 (sometimes written p_0). Use this hypothesized value under the square root, not $\hat{p}$.

Key takeaways

- Test statistic = (sample value − hypothesized value) ÷ standard error.
- Mean: use t with s when $n < 30$ and σ is unknown. Otherwise use z.
- Proportion: $z = (\hat{p} - p)/\sqrt{pq/n}$, with p from H_0.
- Compare with the critical value(s) to reject or fail to reject H_0.